Tigran Khachaturyants, FRS
Curriculum Vitae

Professional Address

Shanghai Jiao Tong University
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Shanghai, PRC

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Education

Ph.D. in Astrophysics, University of Central Lancashire, 2017–2021

Ph.D. Thesis Topic: The influence of warps on the vertical structure of galactic discs
Advisor: Victor P. Debattista

Specialist in Astronomy, Moscow State University, 2009–2015

Specialist Thesis Topic: The Role of Global Spiral Patterns in Star Formation
Advisor: Anatoly V. Zasov

Research Interests

Galactic and extragalactic astrophysics

Vertical structure of galactic discs: warps, bending and breathing waves

Dynamics of spiral galaxies (spiral density waves, bar dynamics, and radial migration)

Structure and dynamics of the Milky Way

Work with astrometric (*Gaia*) and ground-based spectroscopic (GALAH, APOGEE, LAMOST) surveys

Publications

Refereed Publications

Borbolato, L., Amarante, J. A. S., Perottoni, H. D., et al. (incl. **Khachaturyants, T.**), The Low-α Splash Population in the Milky Way, *The Astrophysical Journal, 1008, L23 (2026)*

Amarante, J. A. S., Laporte, C. F. P., Debattista, V. P., et al. (incl. **Khachaturyants, T.**), GASTRO Library. II. Exploring Chemical Bimodalities in Disk Galaxies with GSE-like Mergers and Massive Star-forming Clumps, *The Astrophysical Journal, 1004(2), 242 (2026)*

Wang, S., Brown, A. G. A., Debattista, V. P., **Khachaturyants, T.**, Phase spirals induced by the gas warp, *Astronomy & Astrophysics, 710, A120 (2026)*

Fiteni, K., Anderson, S. R., Debattista, V. P., Caruana, J., Amarante, J. A. S., Gough-Kelly, S., Eyer, L., Beraldo e Silva, L., **Khachaturyants, T.**, Cuomo, V., The edge of the Milky Way's star-forming disc: Evidence from a 'U-shaped' stellar age profile, *Astronomy & Astrophysics, 708, A252 (2026)*

Debattista, V. P., **Khachaturyants, T.**, Amarante, J. A. S., Carr, C., Beraldo e Silva, L., Laporte, C. F. P., Azimuthal metallicity variations, spiral structure, and the failure of radial actions based on assuming axisymmetry, *Monthly Notices of the Royal Astronomical Society, 537(2), 1620–1645 (2025)*

Huang, Y., Feng, Q., **Khachaturyants, T.**, Zhang, H., Liu, J., Shen, J., Beers, T. C., Lu, Y., Wang, S., Yuan, H., A slightly oblate dark matter halo revealed by a retrograde precessing Galactic disk warp, *Nature Astronomy, 8(10), 1294–1301 (2024)*

Garver, B. R., Nidever, D. L., Debattista, V. P., Beraldo e Silva, L., **Khachaturyants, T.**, Exploring the Evolution of Massive Clumps in Simulations That Reproduce the Observed Milky Way α-element Abundance Bimodality, *The Astrophysical Journal, 953(2), Article 128 (2023)*

Debattista, V. P., Liddicott, D. J., Gonzalez, O. A., Beraldo e Silva, L., Amarante, J. A. S., Lazar, I., Zoccali, M., Valenti, E., Fisher, D. B., **Khachaturyants, T.**, Nidever, D. L., Quinn, T. R., Du, M., Kassin, S., The imprint of clump formation at high redshift. II. The chemistry of the bulge, *The Astrophysical Journal, 946(2), Article 118 (2023)*

Khachaturyants, T., Debattista, V. P., Ghosh, S., Beraldo e Silva, L., Daniel, K. J., The pattern speeds of vertical breathing waves, *Monthly Notices of the Royal Astronomical Society: Letters, 517(1), L55–L59 (2022)*

Khachaturyants, T., Beraldo e Silva, L., Debattista, V. P., Daniel, K. J., Bending waves excited by irregular gas inflow along warps, *Monthly Notices of the Royal Astronomical Society, 512(3), 3500–3519 (2022)*

Ghosh, S., Debattista, V. P., **Khachaturyants, T.**, Age dissection of the vertical breathing motions in Gaia DR2: evidence for spiral driving, *Monthly Notices of the Royal Astronomical Society, 511(1), 784–799 (2022)*

Khachaturyants, T., Beraldo e Silva, L., Debattista, V. P., How stars formed in warps settle into (and contaminate) thick discs, *Monthly Notices of the Royal Astronomical Society, 508(2), 2350–2369 (2021)*

Debattista, V. P., Liddicott, D. J., **Khachaturyants, T.**, Beraldo e Silva, L., Box/peanut-shaped bulges in action space, *Monthly Notices of the Royal Astronomical Society, 498(3), 3334–3350 (2020)*

Beraldo e Silva, L., Debattista, V. P., **Khachaturyants, T.**, Nidever, D. L., Geometric properties of galactic discs with clumpy episodes, *Monthly Notices of the Royal Astronomical Society, 492(4), 4716–4726 (2020)*

Earp, S. W. F., Debattista, V. P., Macciò, A. V., Wang, L., Buck, T., **Khachaturyants, T.**, Drivers of disc tilting I: Correlations and possible drivers for Milky Way analogues, *Monthly Notices of the Royal Astronomical Society, 488(4), 5728–5738 (2019)*

Teaching Experience

Guest Lecturer: Invited lecture within a graduate course, Shanghai Jiao Tong University, 2022

Laboratory Demonstrator: AA1003 *Astronomy* undergraduate course, University of Central Lancashire, 2022

Substitute Lecturer: AA3853 *Cosmology and Galaxies* undergraduate course, University of Central Lancashire, 2020–2021

Course Trainer: Machine Learning in Galactic Modelling, Tuorla Observatory Training School, Turku, Finland, 2018

Teaching Assistant and Tutor: AP2235 *Scientific Computing* undergraduate course in the scientific use of Python, University of Central Lancashire, 2017–2019

Tutoring: Basic Astronomy for primary and high school students, Sternberg Astronomical Institute, 2010–2012

Work Experience

Shanghai Jiao Tong University, China, 2022–2026.

Postdoctoral Fellow

University of Central Lancashire, United Kingdom, 2021–2022.

Research Assistant in Galactic Astronomy

University of Central Lancashire, United Kingdom, 2017–2019.

Teaching Assistant, *Scientific Computing* undergraduate course in the scientific use of Python

Software

POGONA — Python library for the analysis of N-body and N-body+SPH simulations. Used by research groups beyond my own.

Grants

2022-2026 – National Science Foundation of China "Young Scientists Fund", 国家基金委青年科学基金项目, No. 12303013, 30万, Role: Principal Investigator.

2022-2024 – The 73rd batch of China Postdoctoral Science Foundation, 第73批面上资助, No. 2023M732250, 8万, Role: Principal Investigator.

2023 – The Shanghai "Super Postdoctoral"Incentive Program, 上海市"超级博士后"激励计划, No. 2023345, 60万 (municipal award with institutional matching)

2023-2024 – 年科技部外国专家青年项目, 10万

Summer/Winter Schools

Tuorla Observatory Training School, Turku, Finland, 2018

Course trainer on the topic of Machine Learning in Galactic Modeling.

Sternberg Astronomical Institute Summer Projects, Moscow, Russia, 2010

Project: Determining stellar variability in catalogued photographic plates.

Project Supervisor: Sergei V. Antipin

Conferences and Workshops

Lorentz Center Workshop "Winding, Unwinding and Rewinding the Gaia Phase Spiral", Leiden, Netherlands, 2025

Project: The role of misaligned accretion on the formation of Phase Space Spirals.

Oxford *Gaia* Workshop, Oxford, United Kingdom, 2019

Project: The distribution of radial and vertical actions in the Gaia Phase Space Spiral.

ESA Summer *Gaia* Sprint, Madrid, Spain, 2018

Project: Crossmatching methods between Gaia and spectroscopic surveys - studying the velocity distributions in the Solar Neighbourhood.

Outreach Experience

European Astrofest, London, United Kingdom, 2020

Exhibitor.

University of Central Lancashire, Preston, United Kingdom, 2018

Moor Park Observatory lecture for the Lancashire Council of Mosques.

Moscow State University, Moscow, Russia, 2010–2015

Lectures on Astronomical Instrumentation during the annual Moscow Science Festival.

Sternberg Astronomical Institute, Moscow, Russia, 2010–2014

Lectures on Observational Astronomy during the annual Astronomical Festival.

Other Considerations

Computer skills

Operating Systems: Windows, macOS, Linux

Languages: C, Python (Galpy, AGAMA, Pynbody), LaTeX

Languages

Native: English, Russian

Basic: Chinese (普通话), French, Armenian